

Profile Summary

I am a software engineer passionate about embodied AI and building intelligent autonomous systems that bridge perception, localisation, planning, and control. My interests span computer vision, state estimation, motion planning, and robust autonomy, backed by a strong mathematical background in stochastic optimisation, signal processing, and control systems. I enjoy turning research ideas into practical software systems.

Skills

Machine Learning & Perception: PyTorch, reinforcement learning, imitation learning, DINOv3, stereo depth estimation

Robotics, Navigation & Control: ROS/ROS2, Gazebo, localisation and mapping, SLAM, A* path planning, obstacle avoidance, trajectory optimisation, MPC, PID control, state estimation, sensor fusion, Kalman filtering, EKF/UKF, stochastic optimal control, robust control

Programming & Software Engineering: Python, C#, .NET, C/C++, JavaScript/TypeScript, Bash, Linux, Docker, FastAPI, PostgreSQL
Cloud, Infrastructure & AI Tooling: Azure, Azure AI Foundry, Semantic Kernel, KVM/QEMU, distributed systems, validation workflows, build/debug tooling, MCP, RAG, knowledge graphs, semantic retrieval, agentic AI workflows, CLI/API integrations

Relevant Work Experience

Software Engineer

Jan 2026 – Present

Microsoft, Melbourne, Australia

- Own and deliver production changes across Azure networking metadata systems including NBaaS, Networking-Metadata, NetGraph, and Monkey Service, improving metadata correctness, validation coverage, buildout reliability, and operational safety across multiple services and repositories.
- Drive device onboarding, schema evolution, and complex issue resolution through scalable onboarding patterns, validation safeguards, and configuration-driven workflows, reducing manual intervention and making infrastructure updates safer and more repeatable.
- Built the NMACopilot agentic memory workstream into a team AI framework spanning VS Code, Web API, CLI, and Copilot integrations, enabling engineers to reuse investigation context across ICM triage, route/IPAM analysis, build-log review, and performance debugging.
- Designed MCP resources/prompts, multi-level memory caching, semantic retrieval, and debugging skills to reduce repeated manual investigation work and make complex networking diagnostics easier to execute across the team.

Software Engineer

Aug 2025 – Nov 2025

MX3 Diagnostics, North Melbourne, Australia

- Researched and designed backend architecture for MX3's consumer diagnostics product, defining scalable, compliance-aware application, data, and service-layer patterns on AWS intended to support millions of users and petabyte-scale multi-source health-data ingestion, analysis, and user-facing insights.
- Helped shape model, infrastructure, and KMM app direction for AI systems that analyse and predict health metrics from smart watches, phones, and MX3 devices, translating early product requirements, privacy constraints, and HIPAA-aligned design considerations into technical architecture, prototypes, and client-facing visualisations.

Research Assistant (C, RF Localisation, Security Protocols, MQTT, React.js, FastAPI, PostgreSQL)

Mar 2025 – June 2025

Department of Electrical and Electronic Engineering, University of Melbourne, Melbourne, Australia

- Built and evaluated an end-to-end real-time RF asset-tracking system for multi-asset localisation, achieving millimetre-level accuracy through RF analysis, wireless security evaluation, and UKF-based sensor fusion across multiple sensors.
- Developed and tested supporting software components across C, MQTT, React.js, FastAPI, and PostgreSQL, enabling stakeholders to visualise real-time tracking behaviour, analyse system outputs, and assess deployment feasibility.

Software Engineering Intern, RADIANT Project (KVM/QEMU VMs, Docker, Linux, Network OS)

Dec 2024 – Mar 2025

Microsoft, Melbourne Australia

- Developed the Anchor Prefix feature for SONiC T2/FRR routing, improving BGP convergence time by 10x and strengthening routing resiliency for large-scale network topologies.
- Built cost-effective multi-T2 virtual topology test environments using cEOS, KVM/QEMU VMs, Docker, Linux, and Network OS tooling, enabling teams to validate resiliency and performance without relying on physical chassis or impacting production devices.
- Delivered technical documentation and internal demos on topology design and validation workflows, supporting knowledge transfer and stakeholder alignment across the team.

Software Engineer Intern (Django, Bash, C, WASM, HTML, CSS)

Mar 2024 – June 2024

Melbourne Telescope Restorative Project, Museums Victoria, Australia

- Developed the Star Map control interface for the Great Melbourne Telescope using Django, Vue.js, and C++, enabling users to select celestial objects and command the telescope to point at them for easier observation.

- Streamlined development and deployment workflows by introducing Docker and Bash automation, making the project easier to set up, reproduce, and maintain for future contributors.

Research Assistant (Kotlin, Objective-C, C, RTOS:ESP-IDF)

Dec 2022 – Jul 2023

Melbourne Defence Enterprise, University of Melbourne, Australia

- Researched and prototyped a secure embedded system for a government client, analysing communication and cryptographic overheads under resource-constrained operating conditions.
- Improved the prototype's practical deployment readiness by optimising throughput, sensor support, and energy efficiency while working within confidentiality and embedded-system constraints.

Publications

- Mukul Chodhary**, Kevin Octavian, SooJean Han, "Efficient Replay Memory Architectures in Multi-Agent Reinforcement Learning for Traffic Congestion Control." IEEE Intelligent Transportation Systems Conference (ITSC), Jul 2024.
- John S Russo, **Mukul Chodhary**, Myrte Strik, Thomas A Shiels, Chin-Hsuan Sophie Lin, Sam E John, David B Grayden, "Feasibility of Using Source-Level Brain Computer Interface for People with Multiple Sclerosis." IEEE EMBC, Nov 2021.

Education

Master of Electrical Engineering (Autonomous Systems)

Feb 2023 – Nov 2025

The University of Melbourne

- Electrical and Electronics Engineering Discipline Award for Capstone project at 2025 FEIT Endeavour Exhibition.
- 6 scholarships and awards worth \$42,500
- Exchange Semester (July 2023-Dec 2023): **Korea Advanced Institute of Science & Technology (KAIST)**

Bachelor of Science (Electrical Engineering Systems)

Feb 2020 – Nov 2022

The University of Melbourne

- 3 scholarships and awards worth \$20,000

Concurrent Diploma in Computer Science

Feb 2022 – Nov 2022

The University of Melbourne

Selected Autonomy, Robotics and Machine Learning Projects

AI for Industry Challenge | Intrinsic and Open Robotics

March 2026 – Present

- Built an end-to-end robotics pipeline for cable insertion using ROS2 and Gazebo, integrating multi-camera perception, DINOv3-based visual features, Kalman-filter pose estimation, and control-policy development under real-time constraints.
- Developing a learned manipulation policy combining a frozen V-JEPA 2.1 encoder, action-conditioned latent prediction, and ACT-style transformer policy head with high-level planning, critic-guided reranking, and episodic memory for replanning.

NeRF re-implementation | [GitHub](#)

Aug 2025 – Aug 2025

- Implemented a Neural Radiance Fields pipeline with volumetric rendering, ray sampling, positional encoding, and MLP-based scene representation for novel-view synthesis.
- Trained and evaluated the model on the Blender Lego dataset using LPIPS, SSIM, and PSNR, comparing reconstruction quality across perceptual and pixel-level metrics.

Learning of dynamical systems from a finite number of closed loop data points | [GitHub](#)

Jul 2024 – June 2025

- Developed a real-time adaptive learning and control framework using the Sign-Perturbed Sums (SPS) algorithm and stochastic robust optimal control, iteratively improving system identification from limited closed-loop data to refine the control law over time with exact probabilistic guarantees.
- Designed a modular asynchronous architecture for integration with real-time controllers in uncertain environments, and built an efficient Python package for discrete-time filtering using Numba, NumPy, and SciPy.

Underwater Stereo Depth Estimation Research | [GitHub](#)

Mar 2024 – Nov 2024

- Developed a self-supervised transfer-learning framework for underwater stereo depth estimation, combining spatial priors from in-air models with stereo deep-learning methods to improve depth prediction in visually degraded underwater environments.
- Designed training and evaluation pipelines across four underwater datasets and achieved state-of-the-art results across eight standard depth-estimation metrics.

Episodic Control with Multi-Agent Reinforcement Learning (MARL) | [GitHub](#)

Jul 2023 – Dec 2023

- Proposed a modular graph-based multi-agent reinforcement learning framework integrating episodic memory control, state-space abstractions, and long-/short-term memory mechanisms to support adaptive forgetting and scalable decision-making.
- Applied the framework to large-scale traffic control, demonstrating strong performance improvements over state-of-the-art MARL baselines; published in IEEE ITSC 2024.

Autonomous Car: ORB-SLAM, Navigation and Path-planning

Jul 2023 – Dec 2023

- Built an autonomous driving stack in ROS integrating ORB-SLAM, global/local planning, MPC control, obstacle avoidance, and trajectory optimisation for closed-loop real-time navigation in dynamic simulation environments.

DNN-LSTM with Hybrid MPC for Home Energy Management System (HEMS) | [GitHub](#)

Mar 2023 – Jul 2023

- Integrated DNN-LSTM forecasting with hybrid MPC to optimise battery storage, heat pumps, and solar curtailment using Bureau of Meteorology and synthetic home-energy data, minimising electricity costs while maintaining stable home temperatures.

Leadership

Technical Lead CereCe BCI-CV Wheelchair Project, MBSI, Melbourne, Australia • Led a 6-8 person team to build an assistive robotic wheelchair platform integrating BCI input, computer vision, perception, networking, A* path planning, PID control, and electromechanical hardware for precise assisted mobility. • Designed and integrated wheelchair and eye-tracking glasses subsystems to improve control accuracy and human-in-the-loop operation; presented the work at the 2024 Australasian Neuroscience Society Conference.	Jul 2022 - Jul 2023
Education Director and Industry Manager, Melbourne University Electrical Engineering Club, Melbourne, Australia • Restructured the organisation by introducing 3 sub-committees, improving knowledge retention, continuity, and scalability across student-led engineering programs. • Coordinated 7-8 technical workshops per semester, information sessions, panels, and industry nights in collaboration with FEIT, the Department of EEE, industry partners, and student organisations. • Ran weekly interactive workshops with students and liaised feedback and concerns to the Department of EEE to improve student support and engagement. • Organised hackathons for 200+ students addressing e-waste sustainability through Watthack and accessibility through Universal Design Sprint.	Sep 2021 - Jul 2023
Education Head Superhack Organising Committee, Superhack, Melbourne, Australia • Relaunched a 2-day hackathon promoting gender diversity in STEM for high school students across Victoria. • Trained 40 university mentors and secured \$8,000 in industry sponsorship to support participant learning and event delivery.	Dec 2022 - Jul 2023